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ITAI-1378 Comp Vision-
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Image Processing Fundamentals

Working through this lab helped me understand image processing in a way that finally clicked. Even though I work with images constantly, traffic camera feeds, before-and-after photos in my med spa and the pictures I edit on my phone, I never really thought about what an image is underneath the surface. Seeing that every image is just a matrix of numbers surprised me more than anything else. It made me realize how much structure and precision exists behind something that feels so visual and intuitive. Separating the RGB channels and converting images to grayscale using different formulas showed me how much control we actually have over the appearance of an image just by manipulating numerical values.

The operations we implemented today helped me connect traditional image processing to the AI tools we talked about in class, especially Nano Banana. When I use AI tools, it feels like magic press button and the image transforms. But this lab showed me that those tools rely on the same mathematical operations I performed manually. Convolution kernels reminded me of how neural networks detect edges and patterns. Histogram equalization felt like a simplified version of how AI models normalize inputs. CLAHE made me think about how AI enhances low-light images. Understanding these foundations changed how I see AI tools; instead of thinking of them as black boxes, I now see them as layered systems built on basic transformations.

Throughout the lab, I kept thinking about real-world applications. In traffic engineering, image enhancement can make roadway cameras more reliable, especially at night or in bad

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weather. Sharpening and contrast adjustments can help detect vehicles or pedestrians more accurately. In my body contouring work, these techniques can improve the clarity of before-and-after photos without altering the truth of the results. Even histogram equalization could help balance lighting differences between photos taken at different times of day. Beyond my personal fields, these techniques are used in medical imaging, satellite analysis, security systems, and smartphone cameras.

The most interesting visual effect came from combining sharpening with a slight contrast increase and a small brightness reduction. It created a dramatic, almost cinematic look that made edges pop while keeping the overall tone rich and balanced. It reminded me of how professional photo editors enhance images to make them look more defined without making them look fake. It also made me think about how these operations could be used to highlight features in traffic images or emphasize contours in body sculpting photos.

For future exploration, I want to dive deeper into segmentation and feature extraction. Enhancement is powerful, but segmentation feels like the next step—actually identifying and isolating meaningful parts of an image. As someone who works with both traffic systems and human body shapes, segmentation could help me understand how AI identifies vehicles, lanes, or anatomical features. It feels like the bridge between traditional image processing and full computer vision.

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Understanding the mathematical foundation behind these operations changed my perspective on AI image tools. Traditional methods give you control and transparency, while AI-powered approaches give you speed and adaptability. Traditional techniques are predictable and easy to explain, but they can be limited when dealing with complex or noisy images. AI tools are powerful and flexible, but they can be harder to interpret and sometimes behave unpredictably. Combining both approaches seems like the best path forward. For example, I could use traditional enhancement to clean up an image before feeding it into an AI model, or use AI to segment an image and then apply traditional filters to highlight specific regions.

This lab helped me understand that image processing is not just a technical skill—it's a language. Once you learn how images are built and how operations transform them, you can communicate with them more effectively. Whether I'm analyzing traffic footage, documenting client progress, or building AI projects, I now have a stronger foundation for understanding and improving the images I work with every day.